

Santhosh Pallepangi

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SKILLS

Programming & Frameworks: Python (pandas, NumPy, scikit-learn, statsmodels), PyTorch, Tensorflow, SQL, SAS, R, Transformers (Hugging Face), LangChain

Machine Learning: Regression, Classification, Bayesian modeling (PyMC), Time series (Prophet/ARIMA), CLV & churn, Uplift/causal, A/B testing, Optimization

GenAI & NLP: Transformers (Hugging Face), RAG, Prompt/Context engineering, LlamaIndex, Langchain, Haystack

Data Warehouses & Eng: Snowflake, Redshift, BigQuery, Databricks (Delta/Spark), dbt, Kafka

Cloud & MLOps: AWS (S3, Glue, Athena, Lambda, SageMaker, Step Functions, ECS), GCP (BigQuery, Vertex AI), Docker, MLflow, Weights & Biases, Git/GitHub Actions, CI/CD, Feature Store

Visualization & BI: Tableau, Power BI, Looker, QuickSight, Plotly, Streamlit, Jupyter, Excel

WORK EXPERIENCE

State of Missouri, *Missouri, US*

Oct 2023 - Present

Data Scientist

- Drove **\$5M** in funding decisions by isolating workforce trends with multivariate regression, Lasso, and XGBoost (CV-validated, SHAP explained)
- Lifted staff retention by 15% in 6 months by predicting turnover risk with logistic regression and enabling targeted, high-ROI HR interventions.
- Reduced forecast error 28% and stockouts 41%, freeing **\$3.2M** in working capital via hierarchical time-series models trained on AWS SageMaker
- Cut model deployment time 80% and tripled experiment throughput by building an AWS-native MLOps stack—S3/Glue/Athena lakehouse, SageMaker, ECR, with CloudWatch drift alarms and Step Functions rollback.
- Enhanced offender classification system accuracy by 18% by training and validating a k-means clustering model on inmate profiles, aiding in facility placement and resource planning

Y Point Analytics, *Hyderabad, India*

May 2018 –July 2021

Sr Data Analyst

- Unified 50+ marketing & sales sources into a governed analytics layer (S3/Redshift), modeled star schemas in dbt, and delivered 99.9% data availability while reducing storage cost 30% via tiered retention.
- Led a 5-analyst BI squad to migrate 1,000+ ELT jobs to dbt + Airflow, cutting dashboard refresh times ~50%, reducing defects ~40%, and publishing a shared data dictionary with business definitions.
- Tripled self-serve analytics adoption and cut ad-hoc requests 45% by launching a governed semantic layer (dbt metrics) and standardized Tableau dashboards.
- Drove +11% marketing ROI by delivering multi-touch attribution and cohort dashboards with common definitions across Sales/Marketing.
- Dropped executive escalations 30% with KPI scorecards and threshold alerts (email/Slack), enabling faster course-corrections.

EDUCATION

University of Missouri, *Columbia, MO*

May 2023

Master of Science in Data Science and Analytics

National Institute of Technology, *Calicut, India*

May 2018

Bachelor of Technology in Computer Science and Engineering

RAG PROJECTS

AI DOC Policy Chatbot

[Try Application](#)

Built RAG chatbot using Gemini Pro to answer policy-related queries from Missouri Department of Corrections documents, integrating PyMuPDF parsing, FAISS vector store, and a Streamlit UI with source citations.

CardioCopilot

[Try Application](#)

Developed a clinical Q&A assistant leveraging the Meditron-7B LLM from Hugging Face, grounded in the Oxford Handbook of Cardiology. Implemented PDF-based semantic retrieval and deployed via Streamlit for real-time domain-specific medical insights.